

From Moduli space to quiver

Let X be a smooth proj CY 3fold
H ample. $\sigma \in H^*(X; \mathbb{Z})$

Let E be a H-polystable sheaf.

then $E = \bigoplus_i E_i^{\oplus a_i}$ E_i : H-stable sheaf.

Associated with this decomposition is

Ext Quiver Q_E $\begin{cases} Q_0 = \{E_1, \dots, E_k\} \\ Q_i: \# \{E_i \rightarrow E_j\} = \dim \text{Ext}^i(E_i, E_j) \end{cases}$

Dimension vector: $\{d_1, \dots, d_k\}$ $\gamma = \sum a_i \text{ch}(E_i)$

Thm (Toda) there exists a superpotential W_E of Q_E s.t

$$\mathcal{M}_X^{\text{H-ss}}(\gamma) \Big|_{V \ni E} \cong \mathcal{M}_{(Q_E, W_E)}(\mathcal{O}) \Big|_U$$

Today: $\mathcal{M}_{(Q, W)}(\mathcal{O})$ for any Q, W and $\mathcal{O}$
and its DT invariant.

Behrend Functions.

suppose $(U, x) \cong (\text{Crit}(f), p)$ for a holomorphic function $f: V \rightarrow \mathbb{C}$

Then. $\chi_B(x) = (-1)^{\dim V} (1 - e(\mathcal{M}_p(f)))$

where $\mathcal{M}_p(f) := \{y \in V: \|y-p\| < \delta, f(y) = f(p) + \varepsilon\}$ $0 < \varepsilon < \delta \leq 1$

Example: $f: \mathbb{C} \rightarrow \mathbb{C}: f(x) = x^{n+1}$ $\text{Cr}(f) = \{0\}$

$$\mathcal{M}_p(f) \cong \bigsqcup_{i=1}^{n+1} \{p_i\}$$

$$\chi_B(0) = (-1)^1 (1 - (n+1)) = n$$

Quiver DT invariants

1. Quiver w/ super-potential:

A quiver is a directed graph:

$$Q = (Q_0, Q_1, s, t)$$

/ | Y
vertices edges $Q_1 \xrightarrow[\text{target.}]{\text{source}} Q_0$

path algebra : $\mathbb{C}[Q] = \{ \text{paths + composition} \} / \text{associativity}$ vertex
= path of length zero
relations of a quiver $I \subseteq \mathbb{C}[Q]$

Example 1 ($\tilde{A}_1$)

$$\begin{array}{ccc} 1 & & 2 \\ \bullet & \xrightleftharpoons[e_2]{e_1} & \bullet \end{array}$$

$$\mathbb{C}[Q] \cong \begin{pmatrix} \mathbb{C} & \mathbb{C}^2 \\ & \mathbb{C} \end{pmatrix}$$

A super-potential of a quiver is an element

$$w \in \mathbb{C}[Q] / [\mathbb{C}[Q], \mathbb{C}[Q]]$$

Notice that : any path without cycles are in $[\mathbb{C}[Q], \mathbb{C}[Q]]$.

$$\text{if: } \gamma = \begin{array}{ccccccc} & e_1 & & e_2 & & & e_{n-1} \\ & \rightarrow & & \rightarrow & & & \rightarrow \\ a_1 & & a_2 & & & & a_n \end{array}, \quad n \neq 1$$

$$\text{then } \gamma = [e_{a_n}, \gamma]$$

Cor : $\mathbb{C}[Q] / [\mathbb{C}[Q], \mathbb{C}[Q]]$ is generated by cycles.

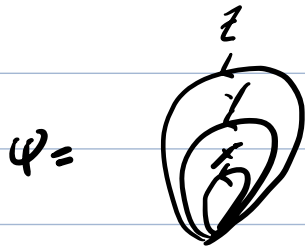
Defn : given $e \in Q'$, define

$$\frac{\partial}{\partial e} : \mathbb{C}[Q] / [\mathbb{C}[Q], \mathbb{C}[Q]] \rightarrow \mathbb{C}[Q]$$

$$\frac{\partial}{\partial x_i} \left(\begin{array}{c} \text{diagram of a cycle } Q_i \text{ with vertices } a_1, a_2, \dots, a_n \text{ and edges } e_1, \dots, e_n \\ a_1 = a_n \end{array} \right) = e_{i+1} \cdots e_n \cdots e_{i-1}$$

$$\text{Def: } \partial W := \left\{ \frac{\partial W}{\partial e} : e \in Q_1 \right\} \subset \mathbb{C}[Q]$$

Example



$$\mathbb{C}[Q] = \mathbb{C}\langle X, Y, Z \rangle$$

$$\text{Let } W = XYZ - XZY$$

$$\text{then } \partial W = \{ [Y, Z], [X, Z], [X, Y] \}$$

$$\mathbb{C}[Q] / \partial W \cong \mathbb{C}[X, Y, Z]$$

2. Moduli space of quiver representations.

A rep of Q is: for each $i \in Q_0$, V_i

for each $e \in Q_1$, $f_e: V_{s(e)} \rightarrow V_{t(e)}$

morphism:

$$\begin{array}{ccc} V_{s(e)} & \xrightarrow{f_e} & V_{t(e)} \\ g \downarrow & & \downarrow g \\ V'_{s(e)} & \xrightarrow{f'_e} & V'_{t(e)} \end{array}$$

Fix $\theta = \{\theta_i\}_{i \in Q_0} \in \mathbb{Z}^{Q_0}$

$$M_Q(\theta) = \left[\prod_{e \in Q_1} \text{Hom}(V_{s(e)}, V_{t(e)}) / \prod_{i \in Q_0} GL(V_i) \right]$$

$$M_Q[\theta] = \prod_{e \in Q_1} \text{Hom}(V_{s(e)}, V_{t(e)}) // \prod_{i \in Q_0} GL(V_i)$$

θ -stability

• Let $\theta = (\theta_i)_{i \in Q_0} \in \mathbb{R}^{Q_0}$ satisfying $\theta(\theta) := \sum_{i \in Q_0} \theta_i \cdot Q_i = 0$

Def. A Q rep (V_i, ρ_e) is θ -(semi)stable if:

for any proper sub rep: $(V_i', \rho_e') \subset (V_i, \rho_e)$, $\theta(\theta') < (\leq) 0$

rmk if $\theta_i \in \mathbb{Z}^{Q_0}$, consider $X_i = \det(\rho_i)^{\theta_i}$

(King) θ -stability $\Leftrightarrow$ GIT stability w.r.t. X

$$\begin{array}{ccccc} M^{\theta\text{-st}} & \hookrightarrow & M^{\theta\text{-ss}} & \hookrightarrow & M \\ \downarrow & & \downarrow & & \downarrow \\ \text{Smooth } M^{\theta\text{-st}} & \rightarrow & M^{\theta\text{-ss}} & \rightarrow & M \end{array}$$

• trace-potential on M

Let $W = \sum (e_i \cdot e_1 \cdots e_n)$

define $\text{Tr}(W) : \mathcal{M}_Q(V) \rightarrow \mathbb{A}^1$

$$\{v, e_i\} \mapsto \sum e_i \cdot \text{tr}(f_{e_n} \circ \dots \circ f_{e_1})$$

• critical locus of $\text{Tr}(W)$

consider $F(X) = \text{Tr}(YX)$

$$\text{then } \frac{\partial F}{\partial X} = Y$$

$\therefore$ critical locus of $\text{Tr}(W) \iff Q$ reps. st. $\mathcal{P}|_{\partial W} = 0$

Example 3



$$W = x^{n+1}$$

$$\partial W = x^n$$

$$\text{Tr}(W)(X) = \text{Tr}(X^{n+1}) = \text{Tr}(X \cdot X^n)$$

$$\text{critical locus} \iff X^n = 0$$

• Br restricting to (Q, W) reps:

$$\mathcal{M}_{(Q, W)}^{\theta\text{-st}}(Q) \subset \mathcal{M}_{(Q, W)}^{\theta\text{-ss}}(Q) \hookrightarrow \mathcal{M}_{(Q, W)}(Q)$$



$$\mathcal{M}_{(Q, W)}^{\theta\text{-st}}(Q) \hookrightarrow \mathcal{M}_{(Q, W)}^{\theta\text{-ss}}(Q) \longrightarrow \mathcal{M}_{(Q, W)}(Q)$$

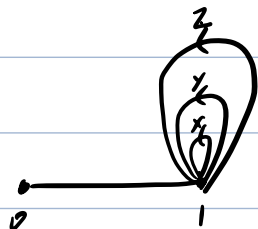
Fact: $\mathcal{M}_{(Q, W)}^{\theta\text{-st}}(Q) = \text{crit}(\text{Tr}(W)) \subset \mathcal{M}_Q^{\theta\text{-st}}(Q)$

Def: If $\theta\text{-ss} = \theta\text{-st}$, then

$$DT_{\theta}(v) = \int_{M_{(Q,W)}^{\theta\text{-st}}(v)} \chi_B \, d\mu$$

where χ_B is the Behrend function.

Example



$$W = XYZ - XZY$$

$$v = (v_0, v_1) = (1, n) \quad \theta = (\theta_0, \theta_1) \text{ s.t. } \theta_0 + n\theta_1 = 0 \quad \theta_1 < 0$$

$$M_{\mathbb{Q}}^{\theta\text{-st}}(v) = M_{\mathbb{Q}}^{\theta\text{-ss}}(v) \cong \text{Hilb}^n(\mathbb{C}^3)$$

$$\text{s.s.} \quad n v'_0 - v'_1 \leq 0 \quad \text{for sub rep}$$

$$\text{if } v'_0 = 1, \text{ then } v'_1 \geq n$$

$$\Rightarrow \text{s.s.} \Leftrightarrow \text{no proper sub-rep can exist with } v'_0 = 1$$

$$\Leftrightarrow \text{let } v = (1, n) \text{ then } \mathbb{C}[X, Y, Z]_v = \mathbb{C}^n$$

$$(X, Y, Z, v) \mapsto \phi: \mathbb{C}[X, Y, Z] \rightarrow \mathbb{C}^n \quad (\Leftrightarrow \text{Ker}(\phi))$$

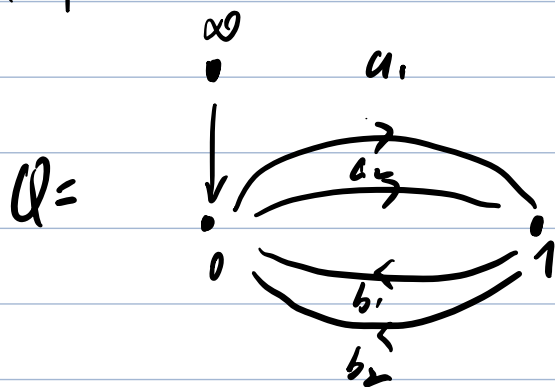
$p \mapsto p \cdot v$

$$DT_{\theta}(1, n) = \int_{\text{Hilb}^n(\mathbb{C}^3)} \chi_B \, d\mu$$

$$\sum_{n \geq 0} DT_{\theta}(1, n) q^n = M(-q) \quad , \quad \text{where } M(q) = \prod_{k \geq 1} \frac{1}{(1 - q^k)^k}$$

$$= 1 + q + 3q^2 + 6q^3 \dots$$

Example



$$W = a_1 b_1 a_2 b_2 - a_1 b_2 a_2 b_1$$

$$X = \text{Tot}(\mathcal{O}_{\mathbb{P}^1}(-1)^{\otimes 2}) \rightarrow \mathbb{P}^1$$

$$\mathcal{O}' = \mathcal{O}_X \oplus \mathcal{O}_X(1)$$

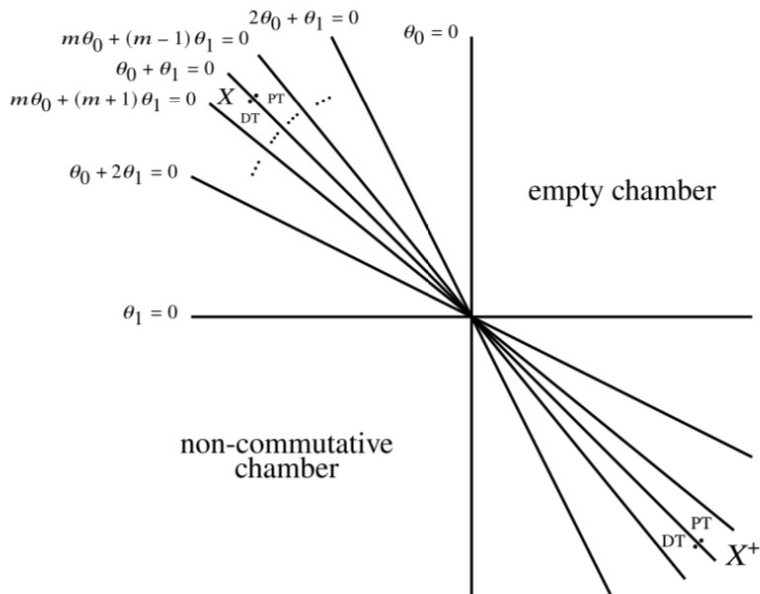
$$\Phi := \text{RHom}(\mathcal{O}', -) : D^b(X) \xrightarrow{\sim} D^b(A)$$

Fact $A \cong \mathbb{C}[QV/\partial W]$

$$Q = (1, m_0, m_1, 2)$$

$$\theta = (\theta_0, \theta_0, \theta_1)$$

Fig. 3.1 Wall-chamber structures



• More on cohomological refinements:

$$; \text{Im}(df: T_V \rightarrow \mathcal{O}_V)$$

Given $f: V \rightarrow \mathbb{C}$ such that $\text{Crit}(f)$ is set theoretically supported on $f^{-1}(0) = V_0$.

One can define a vanishing cycle sheaf $\phi_f \in \text{Perv}(V_0) \subset D_{\text{con}}^b(V_0)$

Fact: $\chi(\phi_f|_x) = (-1)^{\dim V} (1 - e(M_f(x))) = \chi_B$

Def. The cohomological DT invariant is $\text{IH}^*(M, \phi_M)$

$$\text{DT}_4(0) = \sum_{k \in \mathbb{Z}} (-1)^k \dim \text{IH}^k(M, \phi_M)$$